

GNSS / GPS Decoding and Analysis from NMEA 0183 Data

Pravin Oli*

Eötvös Loránd University, IFROS, Hungary

Email: pravin.oli.08@gmail.com

Abstract—This work presents a systematic analysis of GPS data recorded using a simple NMEA 0183 receiver. Two datasets were collected, one stationary and one moving. We evaluate GPS tracks, instantaneous speed, cumulative displacement, precision, HDOP-based error, and satellite statistics to assess GPS performance under stationary and dynamic conditions.

I. INTRODUCTION

Global Navigation Satellite System (GNSS) receivers provide position data with varying accuracy and precision. Analyzing raw NMEA 0183 data helps understand GPS performance, precision under stationary conditions, and variability during motion. This study processes two datasets: `nmea-test1-1.txt` (stationary) and `nmea-test2-1.txt` (moving).

II. DATA PREPARATION

The raw NMEA files were converted to CSV using [NMEA Analyser](#). Key fields include Date, Time UTC, Latitude, Longitude, Altitude, Fix Quality, HDOP, and Active Satellites. The CSV data were cleaned by:

- Removing rows with missing latitude or longitude,
- Converting coordinates to float,
- Combining Date and Time UTC into datetime,
- Resetting indices for consistent processing.

III. MOVEMENT DETECTION AND DISPLACEMENT

Instantaneous speed between consecutive GPS points was computed using geodesic distance. Stationary periods were identified using a speed threshold (< 0.5 m/s). Cumulative displacement was calculated from the starting point.

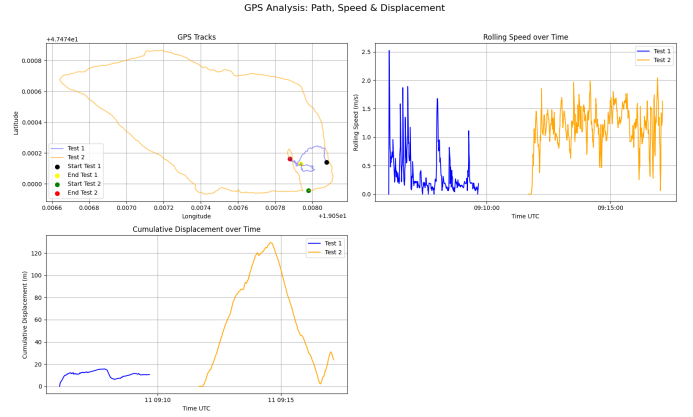


Fig. 1. GPS Analysis: Path, Speed & Cumulative Displacement

IV. GPS PRECISION

A. Stationary Precision (Test 1)

GPS repeatability is measured as standard deviation of coordinates during stationary periods:

$$\sigma_{lat} = \text{std}(\text{latitude}), \quad \sigma_{lon} = \text{std}(\text{longitude}) \quad (1)$$

$$\text{lat}_m = \sigma_{lat} \cdot 111,000, \quad \text{lon}_m = \sigma_{lon} \cdot 111,000 \cdot \cos(\text{mean latitude}) \quad (2)$$

Resulting precision: ~ 5.13 m (latitude), ~ 4.07 m (longitude).

B. Rolling Precision (Test 2)

A moving window (10 points) standard deviation captures short-term GPS variability:

$$\text{LatPrec}_i = \text{std}(\text{lat}_{i:i+9}) \cdot 111,000 \quad (3)$$

$$\text{LonPrec}_i = \text{std}(\text{lon}_{i:i+9}) \cdot 111,000 \cdot \cos(\text{mean latitude}) \quad (4)$$

GPS Precision: Stationary vs Rolling

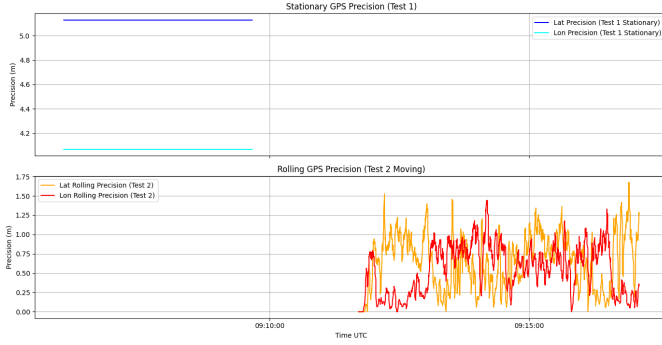


Fig. 2. GPS Precision: Stationary vs Rolling

V. HDOP-BASED HORIZONTAL ERROR

Horizontal Dilution of Precision (HDOP) provides a GPS quality estimate:

$$\text{Error (m)} = \text{HDOP} \times \text{Nominal Error (m)} \quad (5)$$

Nominal error assumed as 5 meters per satellite.

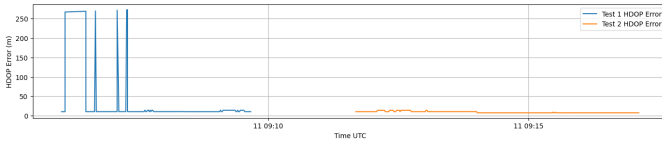


Fig. 3. HDOP-Based Expected Horizontal Error

VI. SATELLITE STATISTICS

Satellite statistics provide further insight into GPS reliability:

- Average in-view satellites: Test 1 = 7.91, Test 2 = 13.99
- Average active satellites: Test 1 = 3.99, Test 2 = 6.57
- Average HDOP: Test 1 = 8.86, Test 2 = 1.88
- Average altitude: Test 1 = 148.92 m, Test 2 = 144.64 m

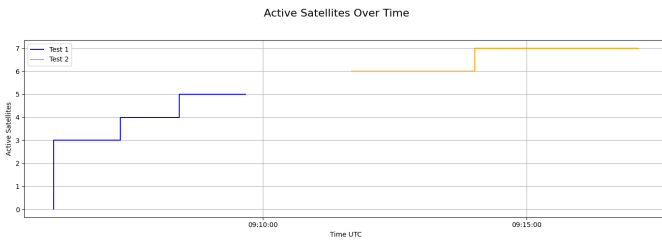


Fig. 4. Active Satellites Over Time

VII. CONCLUSION

- Test 1 is stationary with low speed and tight GPS cluster; precision ~ 5 m.
- Test 2 is moving with higher speed; precision fluctuates due to motion.
- HDOP-based error estimates align with observed precision.
- Satellite statistics confirm GPS quality and reliability.

VIII. QUESTIONS AND ANSWERS

A. Q1: Movement Detection

Question: One of the files was recorded while I was moving, and the other while I was standing still. Which file corresponds to the moving condition, and which one to the stationary condition? Where was I standing still, and where was I moving (describe “what I did” based on the data)?

Answer: Based on the GPS analysis:

- `nmea-test1-1.txt` corresponds to the stationary condition (low speed, tight GPS cluster, rolling precision ~ 5 m).
- `nmea-test2-1.txt` corresponds to the moving condition (speed consistently > 0.5 m/s, cumulative displacement increasing, larger GPS spread).
- Observations: The user was standing still during Test 1, likely at a fixed location, and moving (walking or circling) during Test 2.

B. Q2: GPS Precision / Accuracy

Question: Estimate the precision and/or accuracy (in meters) of the GPS data.

Answer: From the GPS data analysis:

- Stationary precision (Test 1): Latitude ~ 5.13 m, Longitude ~ 4.07 m.
- Moving precision (Test 2): Moving segments show fluctuations deviations (~ 1.5 m) due to actual motion.
- HDOP-based estimated horizontal error aligns with these observed deviations.

C. Q3: Additional Information Extracted

Question: Identify any additional information that can be extracted from the two NMEA data files.

Answer: The NMEA CSV data allows extraction of:

- Number of in-view satellites (Test 1: 7.91, Test 2: 13.99 average)
- Number of active satellites (Test 1: 3.99, Test 2: 6.57 average)
- HDOP values indicating GPS signal quality (Test 1: 8.86, Test 2: 1.88 average)
- Altitude readings (Test 1: 148.92 m, Test 2: 144.64 m)
- Stationary vs moving periods can be automatically detected using speed thresholds.

IX. RESOURCES

The datasets, processed CSV files, Colab notebook, and results for this work are publicly available as follows:

- Raw NMEA files:
 - `nmea-test1-1.txt`
 - `nmea-test2-1.txt`
- CSV files generated using [NMEA Analyser](#):
 - `nmea-analyser-details1-1.csv`
 - `nmea-analyser-details2-1.csv`

- Google Colab notebook:
 - [GNSS / GPS Decoding and Analysis.ipynb](#)
- Report and Resulting figures and plots: [Drive Folder](#)